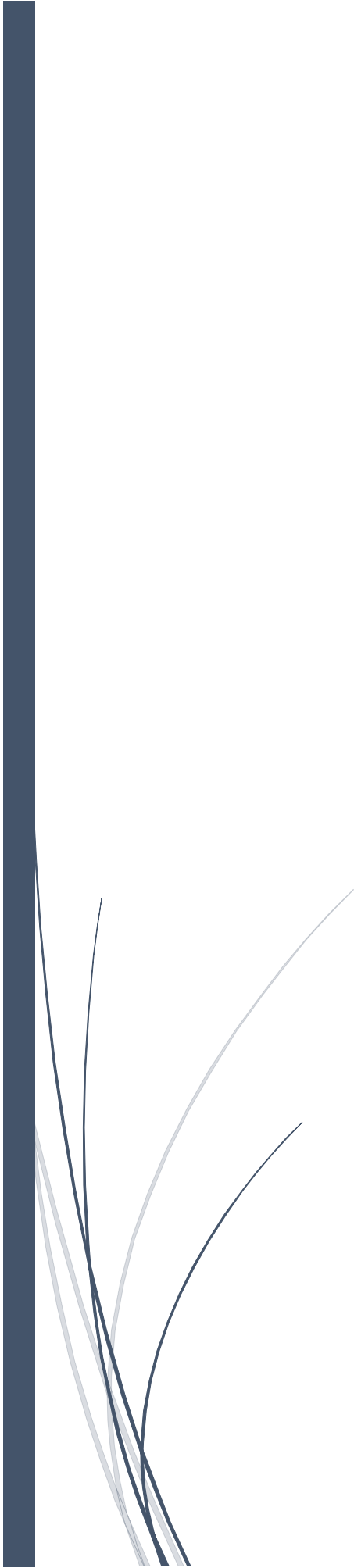


A thick dark blue vertical bar is positioned on the left side of the page. From the bottom of this bar, several thin, curved lines in shades of blue and grey extend upwards and outwards, creating an abstract, organic shape.

X2-Capacitor selection Guideline

How to use the sheet

Abdallah Ebraheem Abdelmotalb Keshk

INPUTS		
Input Parameters	Value	Note
Required Attenuation	40	The required attenuation for the Differential mode noise
Leakage inductance of the Common choke	660	Leakage inductance of the Common choke in μH
Switching frequency	60	Switching Frequency of the SMPS IC IN KHZ
Automatic Equations calculation		
Corner frequency	6	in KHZ (Automated)
X2 Capacitor value	1.07	In μF
Automatic attenuation calculation for selected X2 Capacitor		
Selected X2 Capacitor value	30	In μF
Corner frequency	1.13	in KHZ (Automated)
attenuation atchived	68.99	in dB (Automated)

- 1- **Required attenuation:** the required attenuation for the differential mode noise to be under the limit of standard (if unkown like you are in the early design phase make it as 40 dB as a rule of thumb set by many Well-known companies in the Field like Wurth electronics

Why 40?

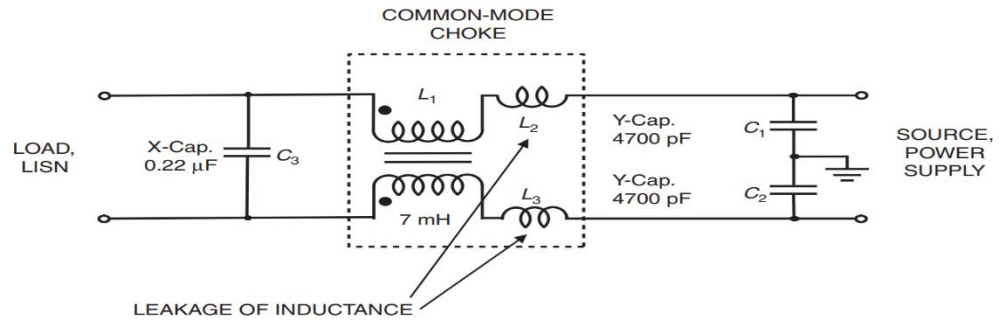
Because it will provide 40dB of attenuation in the Fsw of power ic and its usually the highest spectrum amplitude in the curve (Good Performance as starting point)

- 2- **Leakage inductance:** is the inductance leaked from the common mode inductor in the EMI filter
IF used and its acted as a differential mode filter suppression

13.3.3 Leakage Inductance

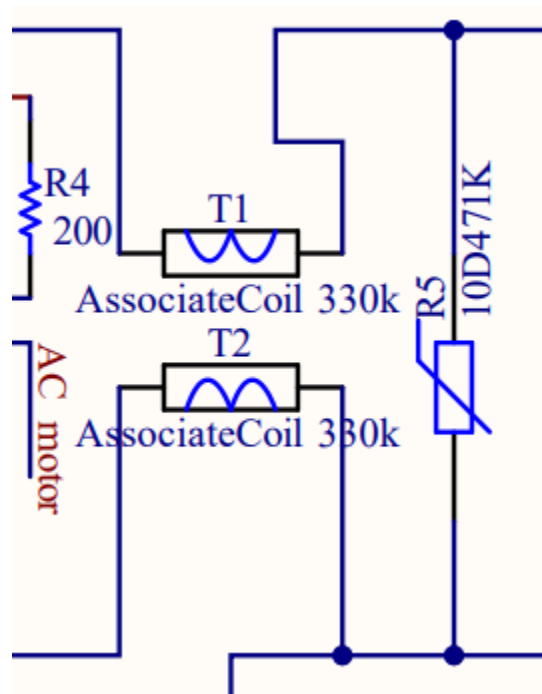
Leakage inductance of the common-mode choke is important in power line filters because it determines the degree of differential-mode inductance present. An ideal common-mode choke provides no differential-mode inductance. The direction of the differential-mode noise current is opposite in each winding, and all the magnetic flux in the core cancels.

The common-mode filter is usually designed first and then the differential mode filter is designed by starting with the leakage inductance of the common-mode choke, and choosing a value for the line-to-line capacitor C_3 , to provide the required attenuation.



Can we use Differential mode inductor in purpose?

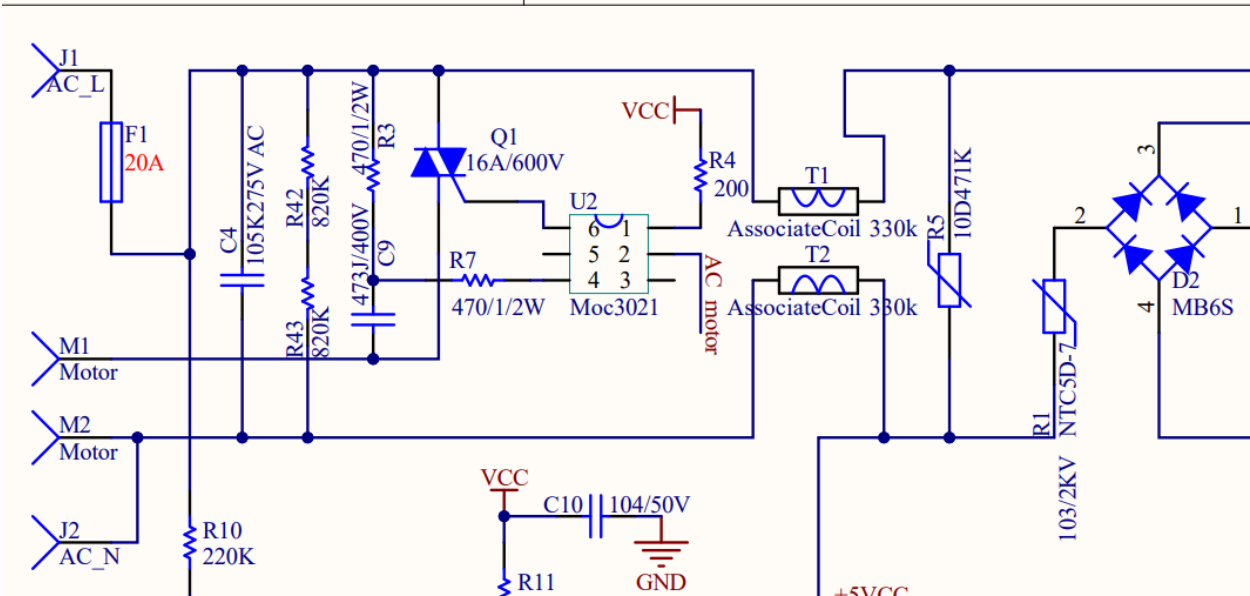
YES, it will be one inductor in the line and the other with Neutral as below



- 3- The switching Frequency: is the frequency of the power IC and can be determined in the power ic datasheet

A real example of the required X2- Capacitor used in Well-known VC brand (Lets use our Excel sheet)

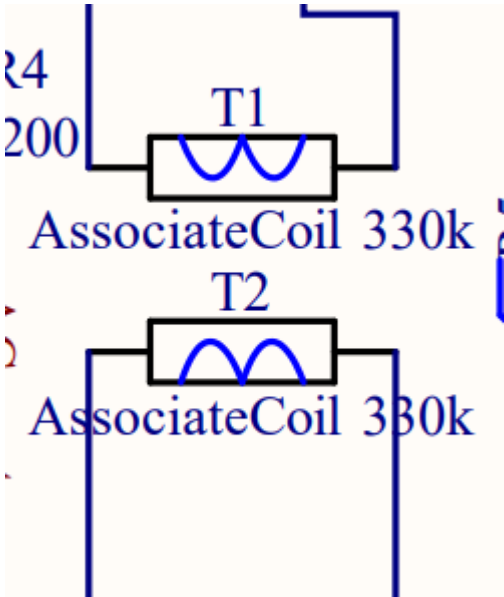
The schematic is here



1- Write the attenuation in db (40 is a good starting point)

INPUTS		
Input Parameters	Value	Note
Required Attenuation	40	The required attenuation for the Differential mode noise
Leakage inductance of the Common choke		Leakage inductance of the Common choke in μH
Switching frequency		Switching Frequency of the SMPS IC IN KHZ
Automatic Equations calaculation		
Corner frequency	0	in KHZ (Automated)
X2 Capacitor value	#DIV/0!	In μF
Automatic attenuation calaculation for selected X2 Capacitor		
Selected X2 Capacitor value		In μF
Corner frequency	#DIV/0!	in KHZ (Automated)
attenuation atchived	#DIV/0!	in dB (Automated)

2-Leakage inductance: here is not specified but he used Differential-Mode inductor as below and the equivalent is 660μH (Because it is Differential) as shown below

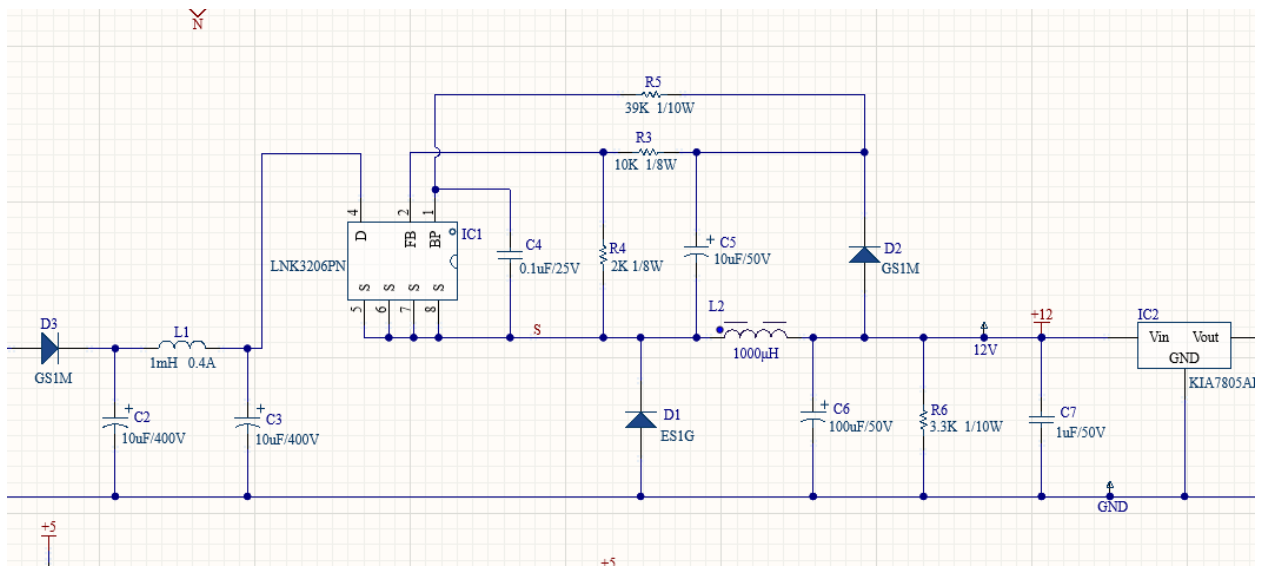


Put it in the sheet as shown:

INPUTS		
Input Parameters	Value	Note
Required Attenuation	40	The required attenuation for the Differential mode noise
Leakage inductance of the Common choke	660	Leakage inductance of the Common choke in μH
Switching frequency		Switching Frequency of the SMPS IC IN KHZ
Automatic Equations calaculation		
Corner frequency	0	in KHZ (Automated)
X2 Capacitor value	#DIV/0!	In μF
Automatic attenuation calaculation for selected X2 Capacitor		
Selected X2 Capacitor value		In μF
Corner frequency	#DIV/0!	in KHZ (Automated)
attenuation atchived	#DIV/0!	in dB (Automated)

Real EMC test proving the theory

Here is the board used in the test



The PWR-IC switch used here is switching at 66KHZ

Product Highlights

Highest Performance and Design Flexibility

- Supports buck, buck-boost and flyback topologies
- Excellent load and line regulation
- Selectable device current limit
- 66 kHz operation with accurate current limit
 - Allows the use of low-cost off-the-shelf inductors
 - Reduces size and cost of magnetics and output capacitor
- Frequency jittering reduces EMI filter complexity
- Pin-out simplifies PCB heat sinking

LET NOW USE THE SHEET AND PLACE THE INPUTS WE COLLECTED

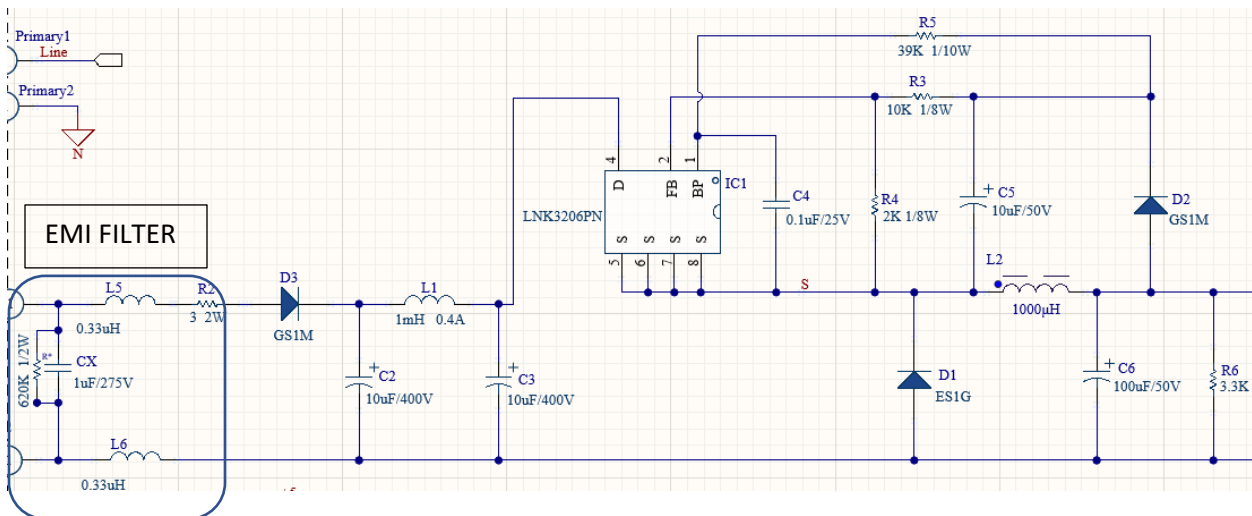
1. Required attenuation is fixed at 40dB if un-known
2. The leakage inductance I don't have a common choke so I used the differential inductors used in hoover VC as mentioned above with value of 330uH
3. The switching frequency of the IC

INPUTS		
Input Parameters	Value	Note
Required Attenuation	40	The required attenuation for the Differential mode noise
Leakage inductance of the Common choke	660	Leakage inductance of the Common choke in μH
Switching frequency	66	Switching Frequency of the SMPS IC IN KHZ
Automatic Equations calaculation		
Corner frequency	6.6	in KHZ (Automated)
X2 Capacitor value	0.88	In μF

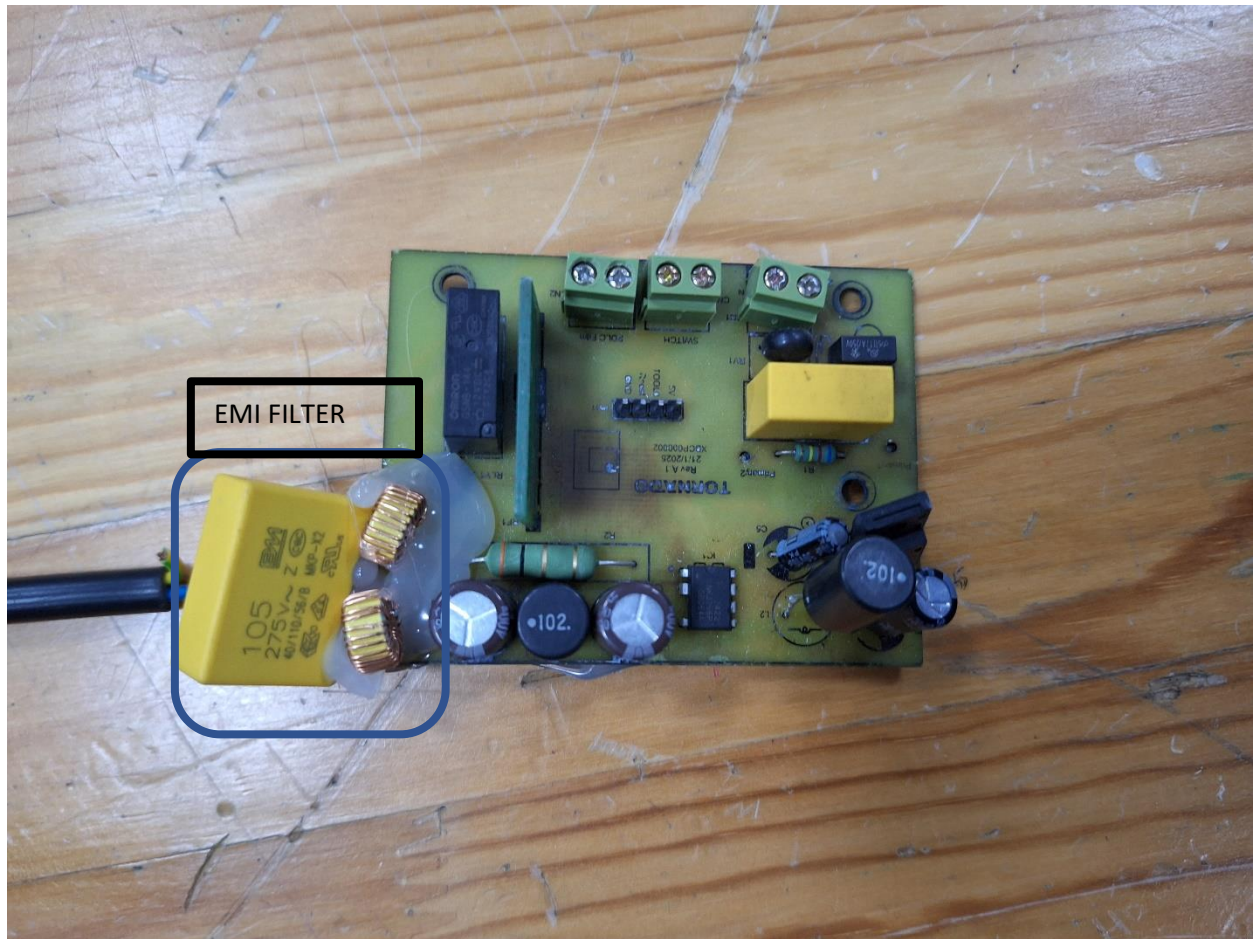
The calculated X2 capacitor for achieving the required attenuation based on the inputs data is 0.88uF

I used the nearest value I have which is 1uF

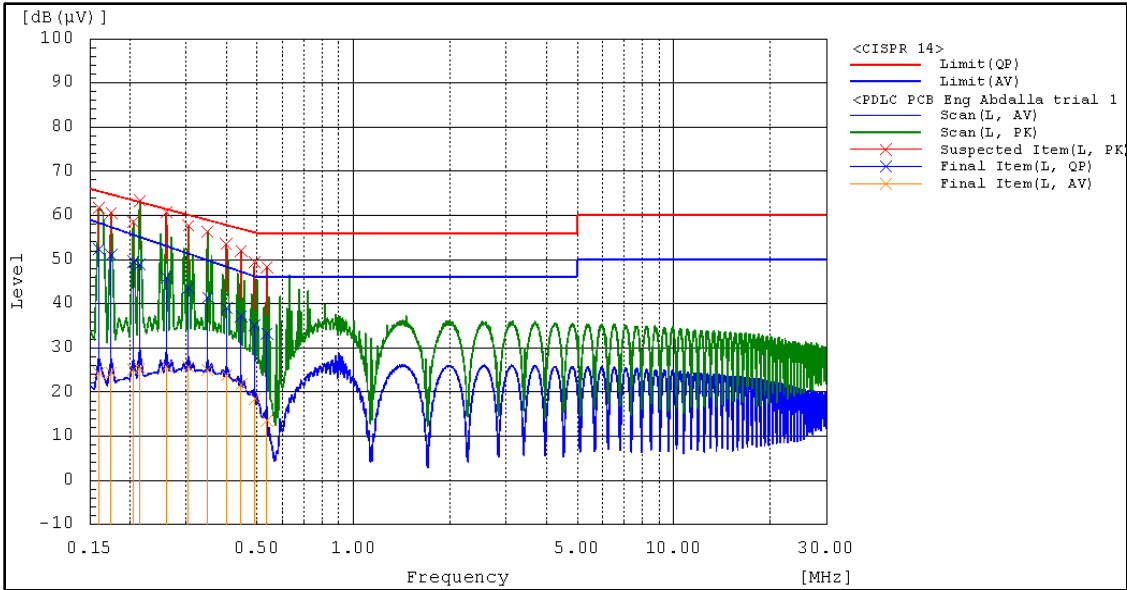
Here is the final circuit



Here is the final board

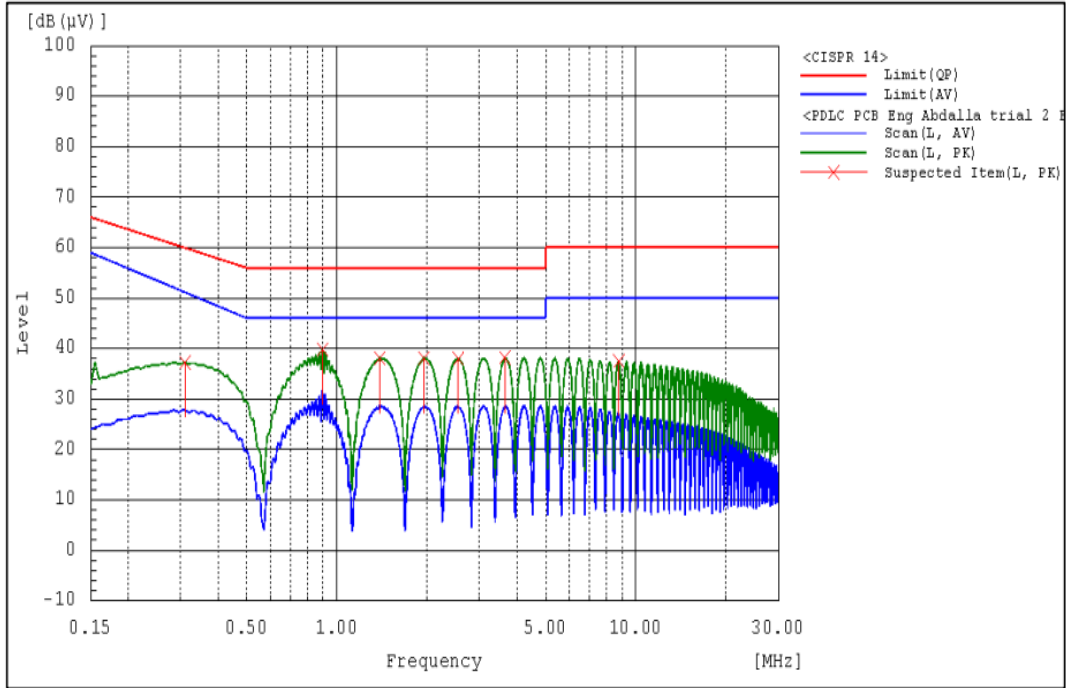


And here is the result Before

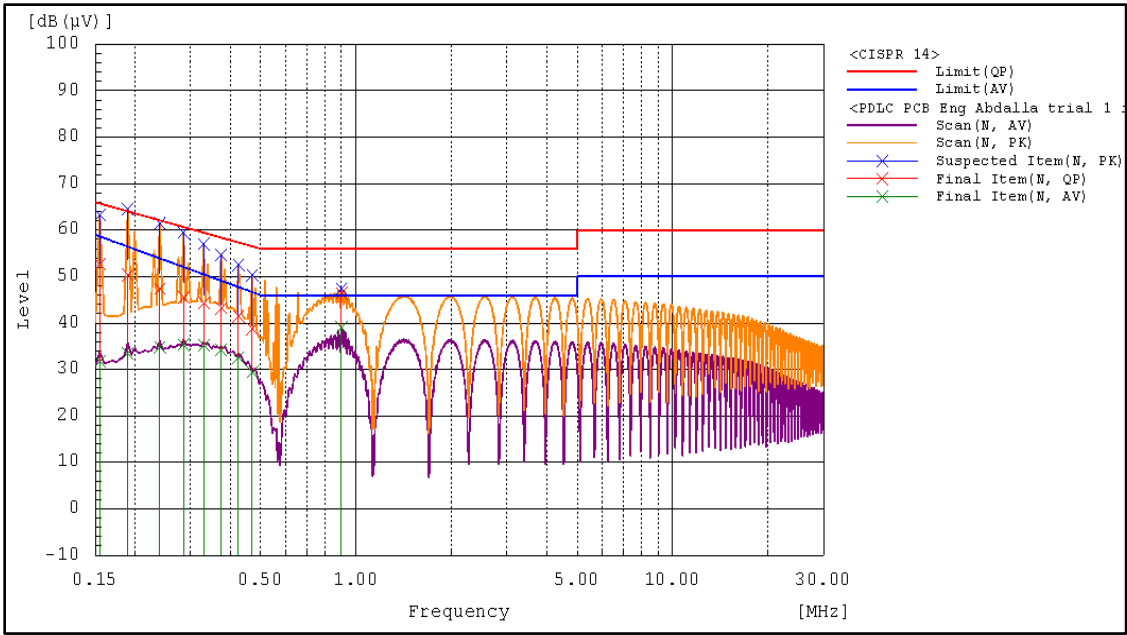


and after with EMI Filter

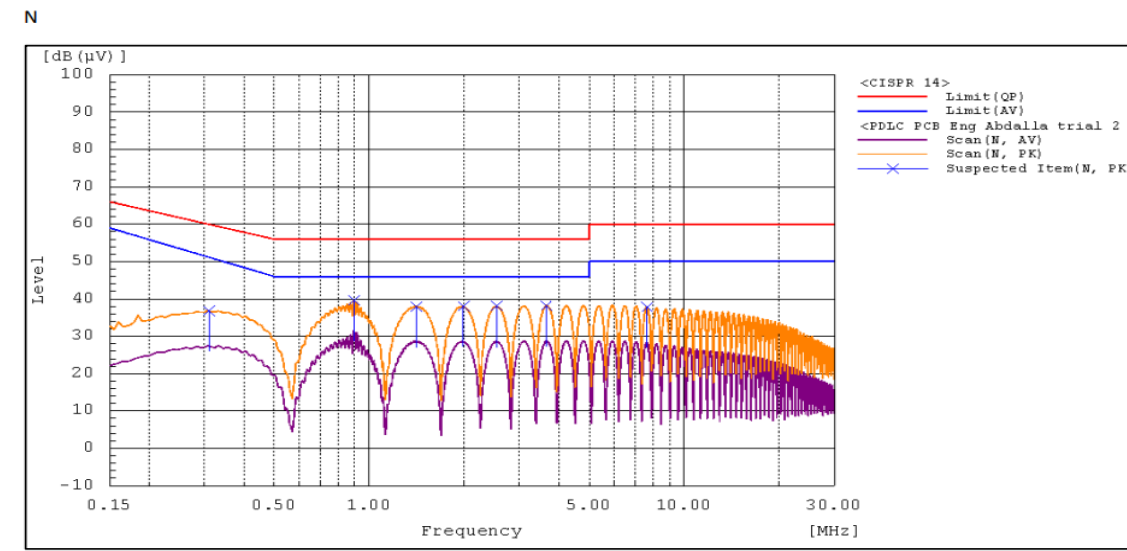
L



Before



After with EMI FILTER



done